

CSA1304-THEORY OF COMPUTATION LAB EXPERIMENTS

4. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$$S \rightarrow 0S0 \mid 1S1 \mid 0 \mid 1 \mid \epsilon$$

AIM

Write a C program to check whether a given string belongs to the Context-Free Grammar (CFG)

CODE

```
#include <stdio.h>
#include <string.h>

int palindrome(char str[], int start, int end)
{
    while(start < end)
    {
        if(str[start] != str[end])
            return 0;
        start++;
        end--;
    }
    return 1;
}

int main()
{
    char str[100];

    printf("Enter the binary string: ");
    scanf("%s", str);

    if(palindrome(str, 0, strlen(str)-1))
        printf("String Accepted\n");
    else
        printf("String Rejected\n");
}
```

```
    return 0;  
}
```

Output

Clear

Enter the binary string: 1001
String Accepted

=== Code Execution Successful ===